

OMRON SYSMAC C200H protocol driver

Overview

The Adroit Omron protocol driver provides serial connectivity using RS-232 or RS-422 to omron host link modules The C200 and C1000 ,CS1 series amongst others, as well as KEYENCE PLC and Process-Plus modules are supported in this driver.

Device configuration

Note: The most commonly used settings are for multi-link system operation with station zero even when on RS232. A baud rate of 19200 is suggested if the default of 9600 does not work. Re-start your agent server after changing any of these settings for them to take effect.

PLC Details

Multi-link System Select to enable multiple link protocol on a RS-422 line. If this option is selected, the Link Unit number must be entered in the 'PLC Station ID' field.

Single-link System Select to disable multiple link (for point to point communication).

PLC Station ID Host Link unit number for multi-link (multidrop) connections

PLC Type Applicable SYSMAC PLC types. The PLC type selected will apply to the primary and secondary port if available. This selection currently has no affect on communications.

Local Details

Port The available serial ports on the machine.

Baud Rate The speed of data transmission. Any of the following Baud Rates may be specified 300, 600, 1200, 2400, 4800, 9600, 19200 or 38400.

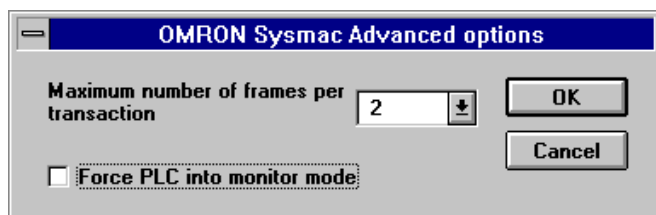
Format The transmission format used for the port.

Retry The number of times that Adroit will try to gain communication with a PLC before failing a transaction.

Time-out The time that Adroit will wait for a valid response from a PLC before failing a transaction (Read or Write).

Advanced dialog

The settings visible in the advanced dialog should only be modified to optimise configuration.



Maximum frames The number of frames that Adroit will grow transactions to. A larger number reduces the number of poll records used, but will reduce performance if the scan points are widely scattered.

Force PLC into monitor mode Allow the Driver to put the PLC into monitor mode on startup as well as whenever the PLC mode is changed to run. If at any stage the driver attempts to write to PLC memory and fails because the PLC is in run mode this option will allow the driver to set the PLC mode while running. The driver will fail to write values to PLC memory if it is in run mode.

The settings in the advanced dialog will be applied to primary and secondary configurations.

Wiring

RS232C to C200HS CPU communications port or C200H-LK201 Host link

CPU Comm port 9 pin Male		PC 9 pin Female
2	————	32
3	————	3
9	————	5
4 + 5 Bridged		4 + 6 Bridged
		8 + 9 Bridged

RS-422 to C200H-LK202 Host link

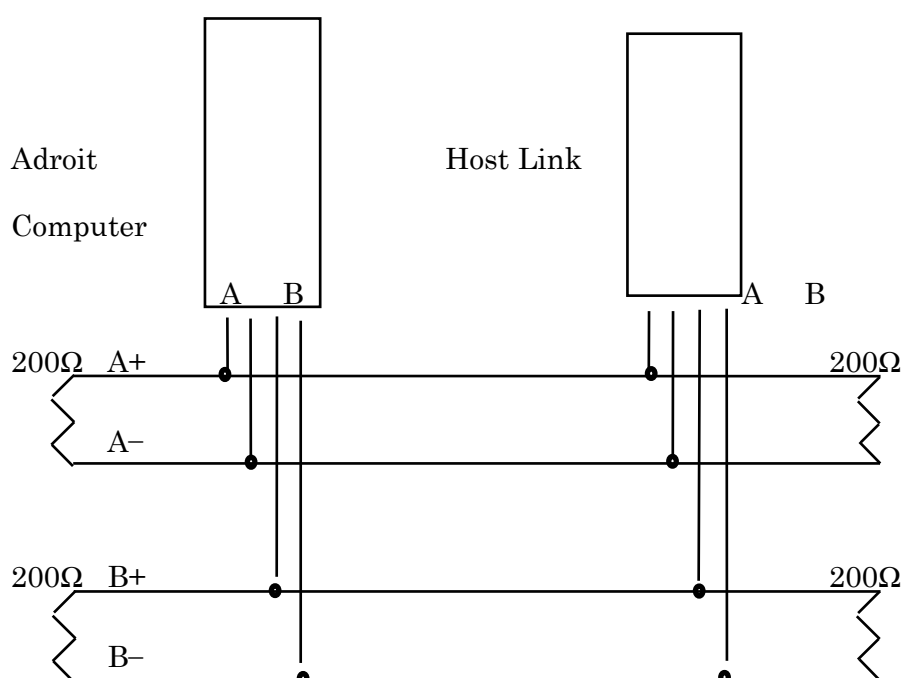
RS-422 port 9 pin Male		RS-422
1	————	RDB (+)
5	————	SDB (+)
3	————	Signal Ground
6	————	RDA (-)
9	————	SDA (-)
7	Screen	Frame Ground

Note: Connect the screen to one side only to prevent earth loops.

The protocol driver was tested using a RS-422 converter and wired as follows:

PC RS-232 port 25 pin Female		RS-422 Converter 25 pin Male	RS-422		Host Link C200H-LK202 RS-422
1	————	1	A+ (TX+)	————	6
2	————	2	A- (TX-)	————	1
3	————	3	Screen	Screen	3
4	————	4	B+ (RX+)	————	5
5	————	5	B- (RX-)	————	9
6	————	6			
7	Screen	7			
8	————	8			
20	————	20			

RS-422 Wiring Diagram

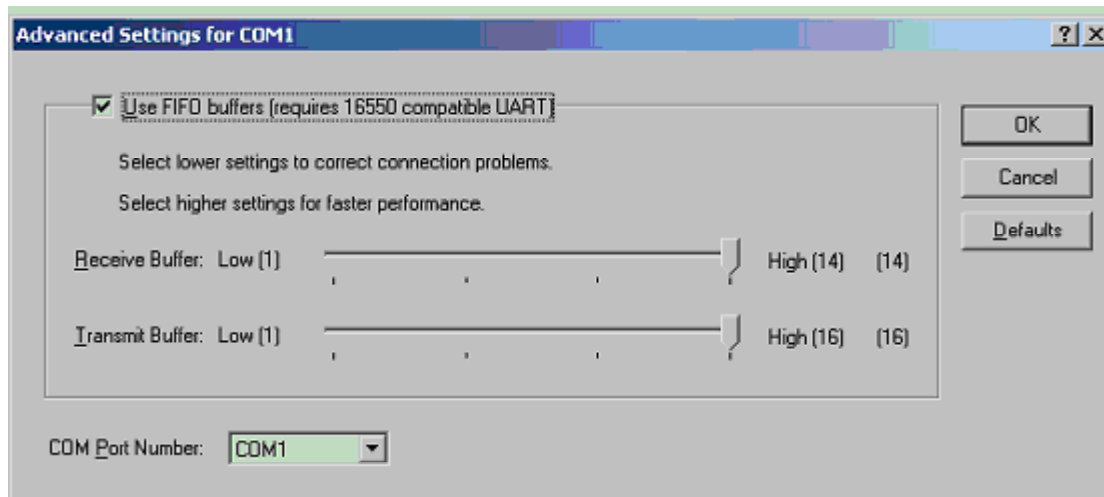


The RS-422 line should be terminated at either end, the total resistance must be 100Ω minimum. The maximum cabling length recommended is 500m. For further details refer to your Host link and RS-422 adapter manuals. Terminating resistance's may not be necessary for cables shorter than 20m.

PC COM Ports – Hardware Setup

If communications appears inconsistent or unreliable, the serial port settings may need some adjusting. Reducing the serial port RX FIFO buffer to 1 (or minimum) may help. If the computer does not have a buffered UART, this setting may not help, in such a case, set the PLC for a lower baud rate.

For Windows 2000 onwards, Right-click on “My Computer”, select manage, then click on device manager. Choose Com & Lpt ports and select the COM port (ie COM1 if you are using COM1). Under the Port settings choose advanced and push the Rx buffer slider to the left most position (ie minimum or 1). Apply changes and the OMRON system should now reply correctly.



Supported addresses

OMRON SYSMAC PLC data addresses supported by Adroit. (* means see note below)

Device	Prefix	Bits	Min Range	Max Range	PLC Data Type	BOOL	INT	REAL	STRING
Internal Relay	IR	1*	000.00	8191.15	Bit	IR 000	IR 000	IR 0000	
		16	000	8191	BCD16		IR 000 B	IR 0000 B	
		16	000	8191	Unsigned Binary (Word)		IR 000 W	IR 0000 W	
		16	000	8191	Signed Binary (Integer)		IR 000 I	IR 0000 I	
		32	000	8190	Unsigned Binary (Double Word)		IR 000 D	IR 0000 D	
		32	00	8190	BCD32		IR 000 N	IR 000 N	
		32	000	4095	Signed Binary (Long)		IR 000 L	IR 0000 L	
Holding Relay	HR	1*	00.00	1599.15	Bit	HR 000	HR 000	HR 0000	
		16	00	1599	BCD16		HR 000 B	HR 00 B	
		16	00	1599	Unsigned Binary (Word)		HR 000 W	HR 00 W	
		16	00	1599	Signed Binary (Integer)		HR 000 I	HR 0000 I	
		32	00	1598	Unsigned Binary (Double Word)		HR 000 D	HR 000 D	
		32	00	1598	BCD32		HR 000 N	HR 000 N	
		32	00	1598	Signed Binary (Long)		HR 000 L	HR 000 L	
Auxiliary Relay	AR	1*	00.00	959.15	Bit	AR 000	AR 000	AR 000	
		16	00	959	BCD16		AR 000 B	AR 000 B	
		16	00	959	Unsigned Binary (Word)		AR 000 W	AR 000 W	
		16	00	959	Signed Binary (Integer)		AR 000 I	AR 000 I	
		32	00	958	Unsigned Binary (Double Word)		AR 000 D	AR 000 D	
		32	00	958	BCD32		AR 000 N	AR 000 N	
		32	00	958	Signed Binary (Long)		AR 000 L	AR 000 L	
Link Relay	LR	1*	00.00	1023.15	Bit	LR 000	LR 000	LR 000	
		16	00	1023	BCD16		LR 000 B	LR 000 B	
		16	00	1023	Unsigned Binary (Word)		LR 000 W	LR 000 W	
		16	00	1023	Signed Binary (Integer)		LR 000 I	LR 000 I	
		32	00	1022	Unsigned Binary (Double Word)		LR 000 D	LR 000 D	
		32	00	1022	Signed Binary (Long)		LR 000 L	LR 000 L	
Timer/ counters (preset)	TCP	16	000	2047	BCD16		TCP 000 B		
		16	000	2047	Unsigned Binary (Word)		TCP 000 W		
		16	000	2047	Signed Binary (Integer)		TCP 000 I		
Timer/ counters (current)	TCC	16	000	2047	BCD16		TCC 000 B		
		16	000	2047	Unsigned Binary (Word)		TCC 000 W		
		16	000	2047	Signed Binary (Integer)		TCC 000 I		
Data Memory	DM	1	0000.00	9999.15	Bit	DM 000			
		16	0000	9999	BCD16		DM 000 B	DM 000 B	
		16	0000	9999	Unsigned Binary (Word)		DM 000 W	DM 000 W	
		16	0000	9999	Signed Binary (Integer)		DM 000 I	DM 000 I	DM 000 S10
		32	0000	9998	Unsigned Binary (Double Word)		DM 000 D	DM 000 D	
		32	0000	9998	Signed Binary (Long)		DM 000 L	DM 000 L	
		32	0000	9998	Float			DM 000 F	
Extended Memory	EM	1	0000.00	9999.15	Bit	EM 000			
		16	0000	9999	BCD16		EM 000 B	EM 000 B	
		16	0000	9999	Unsigned Binary (Word)		EM 000 W	EM 000 W	
		16	0000	9999	Signed Binary (Integer)		EM 000 I	EM 000 I	EM 000 S10
		32	0000	9998	Unsigned Binary (Double Word)		EM 000 D	EM 000 D	
		32	0000	9998	Signed Binary (Long)		EM 000 L	EM 000 L	
		32	0000	9998	Float			EM 000 F	
		32	0000	9998	IEEE754Float			EM 000 R	

- Bit writes are done using a Read-Mask-Write algorithm to achieve an effective bit write. The Read-Mask-Write algorithm reads a word from

PLC memory, masks in the bit to be written and immediately writes the entire word to PLC memory. Therefore if the PLC program modifies the bits between the *Adroit* drivers read and subsequent write operations, the changes made by the PLC program will be lost.

- Strings are scanned as DM 0000 S10 – where S10 is the amount of characters in the string.

Registry keys

Word32Order {DWORD 0 – default do nothing, 1 – swap}

(HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\ProtocolDrivers\OMRSMC\BOILER1\Primary\Word32Order)

Use this registry key to swap the word order for all 32bit types , all these types are affected.

Other OMRON-Protocol PLCs

KEYENCE

Settings to use are 9600, 8 data and even parity. The protocol is the same.

Process Plus

Process Plus modules are supported, an instruction leaflet is provided with the modules on how to configure devices to operate with the I/O modules.

Message protocol

The message format for multiple and single connections must be seen as a wrapper around the messages detailed below. (x10) is used to denote numbers in base 10, (x16) is used to denote numbers in HEX.

Message format for multiple connection protocol:

The message format below should be applied if multiple link selection is made in the Driver setup dialog.

Unit no.		Header and text (see below for contents)		FCS		Terminator	
@	0	0	Header	Text	X	X	* <CR>
FCS calculation range							

Message format for single link protocol:

Header and text (see below for contents)		FCS		Terminator	
Header	Text	X	X	*	<CR>
FCS calculation range					

IR AREA READ:

Send Telegram:

Header	Beginning word (x10)	Number of words (x10)
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R	R	0000	0002
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Reply Telegram:

Header		Response code (x16)		Data from beginning word (x16)	Data from second word (x16)
R	R	0	0	000A	000B

IR AREA WRITE:

Send Telegram:

Header		Beginning word (x10)	Data for beginning word (x16)	Data for beginning word (x16)
W	R	0000	000A	000B

Reply Telegram:

Header		Response code (x16)	
R	R	0	0

HR AREA READ:

Send Telegram:

Header		Beginning word (x10)	Number of words (x10)
R	H	0000	0002

Reply Telegram:

Header		Response code (x16)		Data from beginning word (x16)	Data from second word (x16)
R	H	0	0	000A	000B

HR AREA WRITE:

Send Telegram:

Header		Beginning word (x10)	Data for beginning word (x16)	Data for beginning word (x16)
W	H	0000	000A	000B

Reply Telegram:

Header		Response code (x16)	
W	H	0	0

AR AREA READ:

Send Telegram:

Header		Beginning word (x10)	Number of words (x10)
R	J	0000	0002

Reply Telegram:

Header		Response code (x16)		Data from beginning word (x16)	Data from second word (x16)
R	J	0	0	000A	000B

AR AREA WRITE:

Send Telegram:

Header		Beginning word (x10)	Data for beginning word (x16)	Data for beginning word (x16)
W	J	0000	000A	000B

Reply Telegram:

Header		Response code (x16)	
W	J	0	0

LR AREA READ:

Send Telegram:

Header		Beginning word (x10)	Number of words (x10)
R	L	0000	0002

Reply Telegram:

Header		Response code (x16)		Data from beginning word (x16)	Data from second word (x16)
R	L	0	0	000A	000B

LR AREA WRITE:

Send Telegram:

Header		Beginning word (x10)	Data for beginning word (x16)	Data for beginning word (x16)
W	L	0000	000A	000B

Reply Telegram:

Header		Response code (x16)	
W	L	0	0

TC READ:

Send Telegram:

Header		Beginning word (x10)	Number of words (x10)
R	C	0000	0002

Reply Telegram:

Header		Response code (x16)		Data from beginning word (x16)	Data from second word (x16)
R	C	0	0	000A	000B

TC WRITE:

Send Telegram:

Header		Beginning word (x10)	Data for beginning word (x16)	Data for beginning word (x16)
W	C	0000	000A	000B

Reply Telegram:

Header		Response code (x16)	
W	C	0	0

DM AREA READ:

Send Telegram:

Header		Beginning word (x10)	Number of words (x10)
R	D	0000	0002

Reply Telegram:

Header		Response code (x16)		Data from beginning word (x16)	Data from second word (x16)
R	D	0	0	000A	000B

DM AREA WRITE:

Send Telegram:

Header		Beginning word (x10)	Data for beginning word (x16)	Data for beginning word (x16)
W	D	0000	000A	000B

Reply Telegram:

Header		Response code (x16)	
W	D	0	0

EM AREA READ:

Send Telegram:

Header		Beginning word (x10)	Number of words (x10)
R	E	0000	0002

Reply Telegram:

Header		Response code (x16)		Data from beginning word (x16)	Data from second word (x16)
R	E	0	0	000A	000B

EM AREA WRITE:

Send Telegram:

Header		Beginning word (x10)	Data for beginning word (x16)	Data for beginning word (x16)
W	E	0000	000A	000B

Reply Telegram:

Header		Response code (x16)	
W	E	0	0

STATUS WRITE:

Used to set the PLC back to monitor mode if configured to do so. (see advanced... configuration dialog box)

Send Telegram:

Header		Mode (x16)	
S	C	0	2

Reply Telegram:

Header		Response code (x16)	
S	C	0	0

Performance

This driver supports the standard performance monitor counters described in the user manual.

Troubleshooting

If at first there is no communication, try setting the system to “multi-link” mode with station address 0 (zero).

Simplified cable diagram with fewer conductors (only use this diagram if you have to)

CPU Comm port 9 pin Male		PC 25 pin Female	PC 9 pin Male
2	_____	3	2
3	_____	2	3
7	] [	4	7
8		5	8
5	_____	7	5

Revisions

Date	Revision
30-Oct-97	Added note about multiple and single link operation.
9-Jun-99	Fixed page size and some document formatting, refernces to station # /link unit # clarified.
10-Jul-01	Keyence PLCS KV-300 uses same protocol
20-Jan-03	Added new I/O Area sizes, and new IEEE754 float type.
22-Aug-03	Revisions by NHP included, thanks
30-Aug-05	Added BCD32 support.
22-Jul-09	Added Extended Memory Area Capabilities